Dr. Kritika Sood

PhD | Materials Science | Renewable Energy

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Academic and Research Background

2019-2026	Ph.D. Materials Science **Panjab University, Chandigarh, India** **Institute of Nano Science and Technology (INST), Mohali** **Thesis title - *STRUCTURALLY ENGINEERED PEROVSKITES BASED COMPLEX METAL OXIDE NANOSTRUCTURES FOR ENERGY RELATED APPLICATIONS***
2013-2018	**BS-MS Integrated (Chemistry)** Shoolini University of Biotechnology and Management Sciences, Solan (H.P.), India

Key Research Areas

- Electrochemistry (Ethanol Oxidation/Urea oxidation/Chlorine evolution process/ Water electrolysis)
- Designing and synthesis of nanomaterials
- Industrial waste upcycling
- Renewable energy technologies
- Heterogeneous catalysis

Research Experience

1. Senior Research Fellow (PhD), Institute of Nano Science and Technology/Panjab University, Mohali | Nov 2021 – Jan 2025
2. Junior Research Fellow (PhD), Institute of Nano Science and Technology/Panjab University, Mohali | Nov 2019 – Nov 2021
3. Project Assistant, Institute of Nano Science and Technology, Mohali | Sep 2018 – Mar 2019
4. INSA-NASI Summer Research Fellow, CSIR-National Physical Laboratory, New-Delhi | June 2017 – July 2017

Conference Talks

- "World Nano Congress on Advanced Science and Technology", 8th to 13th March 2021, VIT, Chennai
- International Conference on, "Nanotechnology for Better Living", 7th to 11th September 2021 - NIT, Srinagar
- Frontiers in Chemical Sciences, 2nd to 4th December 2022 – IIT Guwahati, Assam
- International Conference on, "Advanced Materials: Properties and Applications", 20th to 24th February, 2023- Goa University, Goa
- 34th Annual General Meeting of MRSI and 5th Indian Materials Conclave, 12th to 15th December, 2023 - IIT (BHU), Varanasi
- International Conference On Materials for Energy, Environment and Healthcare, December 20th & 21st , 2024 - NIT Calicut

Awards

- Invited as a visiting scholar at Ruhr-Universität Bochum, Germany
- Best Poster Award at Nanotechnology for Better Living at NIT Srinagar
- Awarded the "INSPIRE fellowship" by the Department of Science and Technology, India
- INSA-NASI summer research fellow at CSIR- National Physical Laboratory, New Delhi
- Exchange student at The University of Suwon, South Korea

Research & Technical Expertise

- Design and stabilization of complex metal oxide nanostrctures
- Nanoparticle synthesis: hydrothermal, solvothermal, co-precipitation, sol–gel, acid digestion, waste upcycling
- Knowledge of XPS, UPS, SEM, TGA/DSC, TEM, Drop shape analyzer, Spin and Sputter coating techniques
- Hands-on training on BET, PXRD, ICP-MS, FTIR, DLS, UV-Vis Spectroscopy, Electrochemical workstation
- Proficient in PowerPoint, MS-word, Chem-draw, Origin
- Languages: Hindi (Native), English (Fluent), German and Korean (Basic)

List of Publications

1. **Kritika Sood**, Nausad Khan, Nitu Rani, K. K. Bhasin, Menaka Jha (2026) (in communication) "Nanostructured Nickel Decorated Manganite-based Perovskite Oxide as an Electrocatalyst towards Urea Oxidation"

2. **Kritika Sood**, Nausad Khan, Sapna Devi, K. K. Bhasin, Menaka Jha (2026) (in communication) "Efficient electro-oxidation of ethanol using Ba@NiO nanostructures"

3. Arushi Arora, **Kritika Sood**, Menaka Jha (2026) (under review) "Understanding an insight of excellent hydrogen production from cobalt molybdenum sulphide flowers assembled from ultrafine sheets"

4. Nausad Khan, Anima Mahajan, **Kritika Sood**, Arushi Arora, Santanu Ghosh, Menaka Jha (2026) (under review) “Excellent field emission from ultrafine vanadium dioxide nanorods derive from environmental friendly process”

5. Anupama Tyagi, **Kritika Sood**, Suraj Purohit, and Anil Kumar, “Pd_7Se_2 nanospheres as electrocatalyst for hydrogen evolution reaction (HER)”. *Dalton Transactions* (2026).

6. Supriya Rana, Krishna K Yadav, **Kritika Sood**, Sushma Kumai, Sapna Devi, Tanya Jamwal, Surinder. K. Mehta Menaka Jha (2026), "Efficient Electrocatalytic Ethanol Oxidation Using Lanthanum Nickel Oxide Nanoparticles Anchored on Graphite Surface" *Chempluschem* (2026)

7. Sushma Kumari, Sapna Devi, **Kritika Sood**, and Menaka Jha. "Design of process for stabilization of ultrafine Iron oxide nanostructures and their application as efficient catalyst for chlorine evolution reaction and water remediation." *Materials Chemistry and Physics* (2026): 132094.

8. Sapna Devi, Sushma Kumari, **Kritika Sood**, Santanu Sarkar, Pratik Swarup Dash, and Menaka Jha. "Efficient urea oxidation from strontium ferrite nanostructures synthesized using iron recovered from waste iron ore slime." *Sustainable Materials and Technologies* (2026): e01889.

9. Supriya Rana, Krishna K. Yadav, **Kritika Sood**, Sushma Kumari, Sujit Kumar Guchhait, Surinder K. Mehta, and Menaka Jha. "Shape-induced manganese iron-based nanostructures loaded onto iron sheet as efficient electrocatalyst for O_2 evolution." *Inorganic Chemistry Communications* 183 (2026): 115745.

10. **Kritika Sood**, Arushi Arora, Aman K. K. Bhasin, Kuldip K. Bhasin, Menaka Jha, "Green Synthesis of NiO Nanoparticles Using Waste Tea Extract and Their Utilization as Potential Electroanode for Ethanol Oxidation" *Energy Technology* (2025), p.2500280

11. **Kritika Sood**, Arushi Arora, Nitu Rani, Aman KK Bhasin, Kuldip K. Bhasin, and Menaka Jha. "Boosting Electro-oxidation Efficiency via Chemically Synthesized $GdCoO_3$/g-C_3N_4 Composite: A Promising Anode for Renewable Energy Systems." *Electrochimica Acta* (2025): 147552.

12. Sapna Devi, Supriya Rana, **Kritika Sood**, Sushma Kumari, and Menaka Jha. "Efficient chlorine production from nanostructured cobalt manganese oxide derived from spent battery." *Separation and Purification Technology* (2025): 133213.

13. **Kritika Sood**, Kuldip K. Bhasin, and Menaka Jha, "Understanding electrochemical chlorine evolution process of ultrafine spinel cobalt oxide nanoparticles" *Bulletin of Materials Science* 47, no. 4 (2024): 1-7.

14. **Kritika Sood**, Ritika Wadhwa, Arushi Arora, Kuldip Kumar Bhasin, and Menaka Jha, "Harnessing $LaCoO_3$ Perovskite as a Sustainable Catalyst for Electrochemical Chlorine Evolution in Dye-Contaminated Saline Wastewater" *ACS Applied Energy Materials* (2024).

15. Nausad Khan, Anima Mahajan, Arushi Arora, **Kritika Sood**, Sushma Kumari, Santanu Ghosh, and Menaka Jha "Nanostructured tungsten trioxide developed from the environmentally friendly green process as a promising cathode for excellent field emission" *Materials Chemistry and Physics* 320 (2024): 129364.

16. **Kritika Sood**, Krishna K. Yadav, Kuldip K. Bhasin, Tamal K. Ghosh, Santanu Sarkar, and Menaka Jha, "Unraveling the behaviour of dual active sites of La_2FeMnO_6 for electrochemical oxygen generation in alkaline conditions." *Journal of Electroanalytical Chemistry* 938 (2023): 117449.

17. Preeti Oswal, **Kritika Sood**, Siddhant Singh, Aayushi Arora, Anurag Bahuguna, Suraj Purohit, and Arun Kumar, "Single source precursor route for the first graphene oxide–Pd_6P nanocomposite: application in

electrochemical hydrogen evolution reaction" *Dalton Transactions* 51, no. 17 (2022): 6537-6542.

18. **Kritika Sood**, Supriya Rana, Ritika Wadhwa, Kuldip Kumar Bhasin, and Menaka Jha, "Mechanistic insights of electrochemical Cl_2 and O_2 generation from lanthanum cobalt manganese oxide" *Advanced Materials Interfaces* 9, no. 25 (2022): 2201138.

19. Supriya Rana, Krishna K. Yadav, **Kritika Sood**, Ankush, S. K. Mehta, and Menaka Jha, "Low-temperature hydrothermal method for synthesis of crystalline Fe_2O_3 and their oxygen evolution performance" *Electroanalysis* 32, no. 11 (2020): 2528-2534.

20. Sunaina, Krishna K. Yadav, Sujit K. Guchhait, **Kritika Sood**, S. K. Mehta, A. K. Ganguli, and Menaka Jha, "Mechanistic insights of enhanced photocatalytic efficiency of SnO_2-SnS_2 heterostructures derived from partial sulphurization of SnO_2" *Separation and Purification Technology* 242 (2020): 116835.

Book Chapters

1. **Kritika Sood,** Vishwajit M. Gaikwad, Anand S. Kakde & Krishna K. Yadav “Future Trends in Nanostructured Semiconductor Devices” Two-Dimensional Nanostructured Semiconductors for Clean Energy Solutions. Cham: Springer Nature Switzerland, 2026. 301-325.

2. Deepak Chauhan, **Kritika Sood**, Venugopala Rao Battula and Krishna Yadav "Chemical Waste-Derived Carbon Nanomaterials." Waste Derived Carbon Nanomaterials. Volume 1. American Chemical Society, 2025. 203-224.

3. Menaka Jha, Arushi Arora, and **Kritika Sood** "Wastewater Treatment Using Nanoadsorbents Derived from Waste Materials" In Waste to Profit, pp. 147-164. CRC Press, 2023.

4. Enza Torino, Deepika Jamwal, **Kritika Sood**, Virender Pratap Singh, Pardeep Singh, and Pankaj Thakur, "Bio-inspired polymer composites: Robust biomedical application podium" In Modified Biopolymers: Challenges and Opportunities, pp. 261-284. Nova Science Publishers, Inc., 2017.

Patents

1. A process for producing La_2FeMnO_6 (LFMO) from iron ore slime and applications thereof: Santanu Sarkar, Supriya Sarkar, Menaka Jha, Sujit Gucchait, **Kritika Sood**, Ankush, Krishna Yadav, (2021) Indian patent Application No. 202131014754, Date of filing: **March 31, 2021**